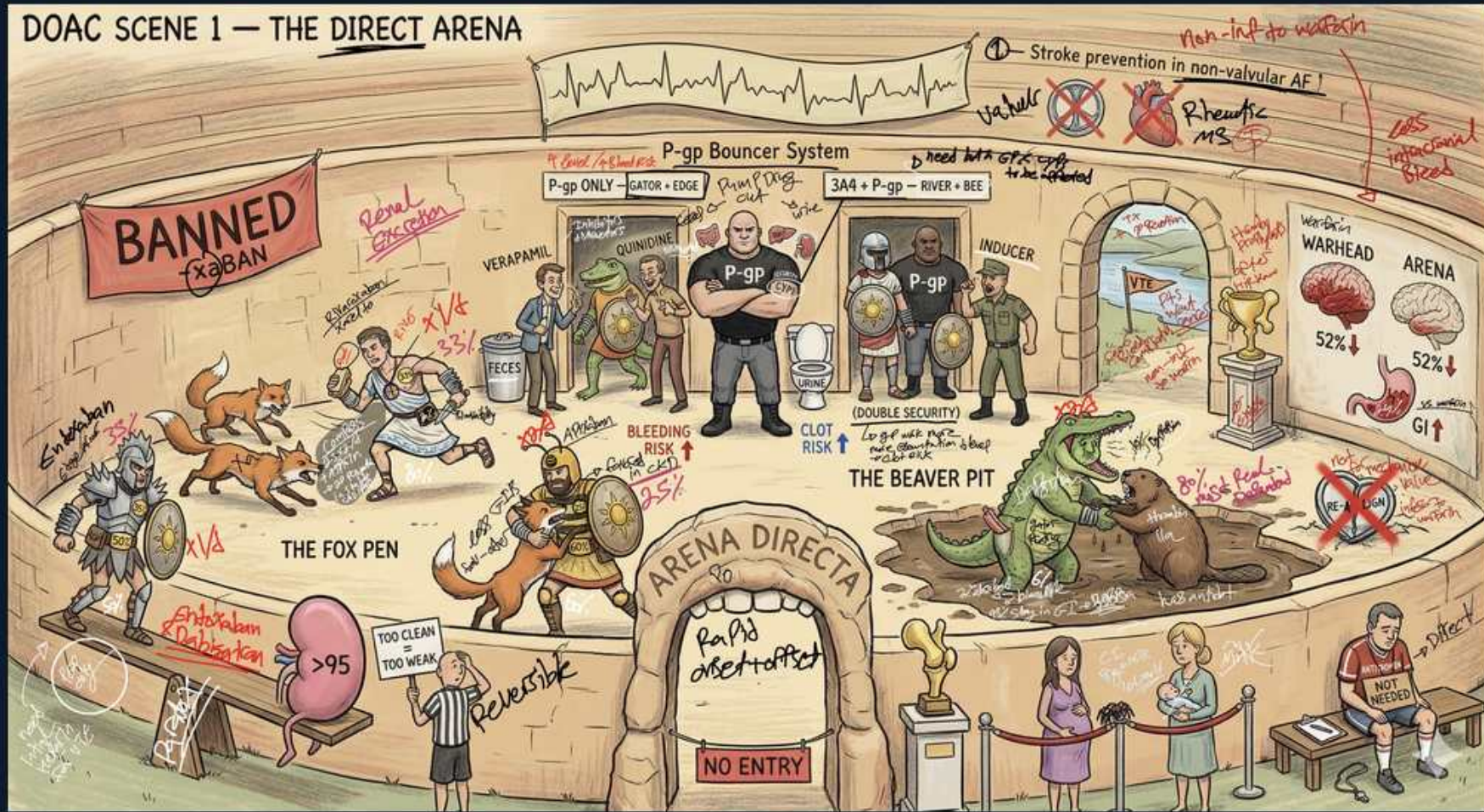


DOAC — The Direct Arena (Mechanisms, Interactions & Contraindications)

DOAC SCENE 1 — THE DIRECT ARENA

1. Direct, rapid onset/offset

WHAT YOU SEE

Central arena gate 'ARENA DIRECTA', 'RAPID DIRECT ONSET/OFFSET', 'NO ENTRY'

WHAT IT ENCODES

DOACs are DIRECT inhibitors of a single clotting factor with RAPID ONSET and OFFSET (peak ~1–4 h, half-life ~8–17 h). This kinetic profile is why they need no bridging and no routine monitoring, and why simply holding the drug is the core of perioperative and minor-bleed management.

BOARD TRAP

Rapid offset is a benefit but also a liability — a missed dose drops anticoagulation quickly, so adherence matters far more than with warfarin. 'Direct' distinguishes them from warfarin (indirect, vitamin-K dependent) and heparin (antithrombin-dependent).

2. Banned: warfarin off-target

WHAT YOU SEE

Red 'BANNED' poster, fox figures, '-xaban' notes

WHAT IT ENCODES

The 'BANNED' wall flags settings where DOACs are NOT allowed and warfarin (or heparin) must be used: mechanical heart valves, moderate-to-severe MITRAL STENOSIS, and antiphospholipid syndrome (especially triple-positive). In these, DOACs showed higher thrombosis/failure rates.

BOARD TRAP

DOACs are contraindicated in mechanical valves and antiphospholipid syndrome — choosing apixaban for a triple-positive APS patient or a St. Jude valve is a classic wrong answer; the correct drug is warfarin.

3. P-gp bouncer system

WHAT YOU SEE

Muscular bouncer 'P-gp', 'P-gp ONLY — GATOR · EDGE'

WHAT IT ENCODES

ALL DOACs are substrates of the P-GLYCOPROTEIN (P-gp) efflux transporter. Dabigatran ('gator') and edoxaban ('edge') depend on P-gp ALONE (they are not significantly CYP-metabolized). Strong P-gp inhibitors raise their levels and bleeding risk; P-gp inducers lower levels and risk clot.

BOARD TRAP

Dabigatran and edoxaban interact via P-gp only; the -xabans (rivaroxaban, apixaban) interact via BOTH P-gp AND CYP3A4 — so a CYP3A4 drug affects riva/apix but not dabigatran. Match the transporter to the drug.

4. Double security: 3A4 + P-gp

WHAT YOU SEE

Two guards 'P-gp', 'DOUBLE SECURITY', '3A4 + P-gp → RIVER · BEE'

WHAT IT ENCODES

Rivaroxaban ('river') and apixaban ('bee') are cleared by BOTH CYP3A4 metabolism AND P-gp transport. Therefore drugs that are dual strong CYP3A4 + P-gp INHIBITORS (e.g., azole antifungals like ketoconazole, HIV protease inhibitors) markedly raise their levels → avoid or reduce dose.

BOARD TRAP

A combined strong 3A4 + P-gp inhibitor (ketoconazole, ritonavir) dangerously elevates rivaroxaban/apixaban — but has less effect on dabigatran (no CYP). Inducers (rifampin, carbamazepine, phenytoin, St. John's wort) lower ALL DOAC levels and risk treatment failure.

5. Verapamil / quinidine — P-gp inhibitors

WHAT YOU SEE

Guards labeled 'VERAPAMIL', 'QUINIDINE', feces icon

WHAT IT ENCODES

P-gp INHIBITORS (verapamil, quinidine, amiodarone, dronedarone, ketoconazole, cyclosporine) increase DOAC levels and bleeding risk — most relevant for dabigatran/edoxaban, which rely on P-gp. Dabigatran is also excreted partly via the gut (feces icon) as well as the kidney.

BOARD TRAP

Verapamil/quinidine/amiodarone raise dabigatran levels (P-gp inhibition) and may require dose reduction — don't assume only CYP drugs matter for DOACs.

6. Bleeding risk (Edge)

WHAT YOU SEE

Figure with shield, 'BLEEDING RISK ↑'

WHAT IT ENCODES

The dominant adverse effect of every DOAC is BLEEDING. Compared with warfarin, DOACs have notably LESS intracranial hemorrhage. The GI-bleed signal varies: dabigatran, rivaroxaban, and high-dose edoxaban have somewhat MORE GI bleeding, while apixaban has the most favorable GI-bleed profile.

BOARD TRAP

DOACs lower intracranial bleeding vs warfarin but several increase GI bleeding — apixaban is the GI-friendliest choice. Don't claim all DOACs reduce all bleeding.

7. Clot risk (toilet / GI)

WHAT YOU SEE

'CLOT RISK ↑', toilet icon near the bouncers

WHAT IT ENCODES

The flip side of strong INDUCERS or under-dosing is CLOT RISK — inadequate drug exposure leads to stroke or VTE recurrence. P-gp/3A4 inducers (rifampin, anticonvulsants, St. John's wort) and inappropriate dose reductions are the main culprits.

BOARD TRAP

Under-dosing a DOAC (e.g., wrongly applying a reduction criterion, or co-prescribing rifampin) causes thrombosis — the board tests both over- and under-anticoagulation directions.

8. Beaver pit = -xabans (Xa)

WHAT YOU SEE

Beaver figures 'THE BEAVER PIT', shields

WHAT IT ENCODES

The BEAVER PIT groups the factor Xa inhibitors — the '-xabans' (rivaroxaban, apixaban, edoxaban). The beaver/X visual anchors that these block factor Xa, contrasted with the gator (dabigatran) that blocks thrombin (IIa). Their antidote is andexanet alfa.

BOARD TRAP

Keep the -xabans (anti-Xa, reversed by andexanet, monitored by anti-Xa assay) separate from dabigatran (anti-IIa, reversed by idarucizumab, monitored by dilute TT).

9. Fox pen = dabigatran/IIa

WHAT YOU SEE

Foxes in a pen 'THE FOX PEN', notes

WHAT IT ENCODES

The fox pen anchors dabigatran's identity as the direct THROMBIN (IIa) inhibitor and the DOAC most renally cleared (~80%) — so it accumulates as CrCl falls and is the one removable by dialysis and reversible with idarucizumab.

BOARD TRAP

Dabigatran is the renal-clearance outlier — avoid it in significant CKD. Its antidote (idarucizumab) and monitoring assay (dilute TT/ecarin) differ from the -xabans.

10. Renal cutoff CrCl >95 too clean

WHAT YOU SEE

'>95 TOO CLEAN', '<... TOO WEAK', referee

WHAT IT ENCODES

Edoxaban carries a unique renal CEILING: it should NOT be used for AF stroke prevention when CrCl is >95 mL/min ('too clean'), because reduced drug exposure led to higher stroke rates. At the other end, severely reduced CrCl ('too weak') contraindicates DOACs due to accumulation.

BOARD TRAP

The 'CrCl too high' contraindication is specific to EDOXABAN — it is the opposite of the usual renal-failure caution and a high-yield distractor.

11. Non-valvular AF indication

WHAT YOU SEE

Top-right heart 'STROKE PREVENTION in non-valvular AF', 'non-valvular', rheumatic crossed

WHAT IT ENCODES

A core indication: STROKE PREVENTION in NON-VALVULAR atrial fibrillation, where DOACs are first-line and at least as effective as warfarin with less intracranial bleeding. 'Non-valvular' specifically EXCLUDES mechanical valves and moderate-severe (rheumatic) mitral stenosis.

BOARD TRAP

'Non-valvular AF' does NOT mean 'no valve disease at all' — it specifically excludes mechanical valves and rheumatic mitral stenosis, where warfarin is mandatory. AF with mild MR/AS can still get a DOAC.

12. VTE trophy / indications

WHAT YOU SEE

'VTE' trophy on pedestal

WHAT IT ENCODES

Beyond AF, DOACs are first-line for acute DVT/PE treatment and extended VTE secondary prevention, and for orthopedic VTE prophylaxis. Cancer-associated VTE can use specific DOACs (apixaban, edoxaban, rivaroxaban) though caution with GI/GU tumors due to bleeding.

BOARD TRAP

In cancer-associated thrombosis with luminal GI/GU tumors, rivaroxaban/edoxaban carry higher GI bleeding — apixaban or LMWH may be preferred. DOACs are not used for catheter/line-related or APS-related clots indiscriminately.

13. Warhead / arena — 52% reductions

WHAT YOU SEE

Brain 'WARHEAD 52%', stomach 'ARENA 52%', 'GI'

WHAT IT ENCODES

Comparative bleeding signal: relative to warfarin, DOACs cut INTRACRANIAL hemorrhage roughly in half (brain 'warhead' ~52% reduction) — their biggest safety advantage — while the GI ('arena/stomach') bleeding picture is drug-dependent and can be higher for some agents.

BOARD TRAP

The headline DOAC benefit is reduced INTRACRANIAL bleeding vs warfarin; do not generalize this to GI bleeding, which can be increased for dabigatran/rivaroxaban/high-dose edoxaban.

14. Pregnancy contraindicated

WHAT YOU SEE

Pregnant figures, crossed-out icons at lower right

WHAT IT ENCODES

DOACs are CONTRAINDICATED in PREGNANCY and breastfeeding (insufficient safety data, placental crossing/fetal risk). The anticoagulant of choice in pregnancy is LMWH, which does not cross the placenta. Warfarin is also teratogenic and generally avoided.

BOARD TRAP

For a pregnant patient needing anticoagulation, the answer is LMWH — not a DOAC and not warfarin (warfarin embryopathy). This is a frequent test pivot.

15. Antithrombin-independent

WHAT YOU SEE

Crocodile/beaver figures with handwritten 'direct' notes near center floor

WHAT IT ENCODES

Unlike heparins (which require ANTITHROMBIN as a cofactor) and warfarin (which depletes vitamin-K-dependent factors over days), DOACs bind and inhibit their target factor DIRECTLY and independently of antithrombin. This gives predictable, antithrombin-independent anticoagulation with fixed dosing.

BOARD TRAP

DOACs work even when antithrombin is low/deficient (direct mechanism) — a contrast with heparin, which fails in antithrombin deficiency. Don't attribute DOAC action to antithrombin.
